Cave Lidar

Engineering Documentation

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1 Overview

A mobile rotating lidar rig for mapping caves. A spinning lidar is rotated about a second axis by a stepper drive, and the sensor pose (IMU) is fused with the lidar data to reconstruct a point cloud. This document describes the hardware, the control software, and the reconstruction and viewing software.

TODO: Project goals, requirements, intended use, system block diagram.

2 Hardware

2.1 Lidar

Velodyne HDL lidar on a 3D-printed mount, rotated about the horizontal shaft axis.

TODO: Exact lidar model, mass, power supply, interface.

2.2 Rotation mechanism

The lidar is mounted on a shaft carried by two 3D-printed bearing brackets on an aluminium T-slot frame. A stepper motor with planetary gearbox drives the shaft through a toothed belt (GT2-type, marked 250-6). A slip ring at the end of the shaft carries power and data to the rotating lidar without limiting the rotation angle.

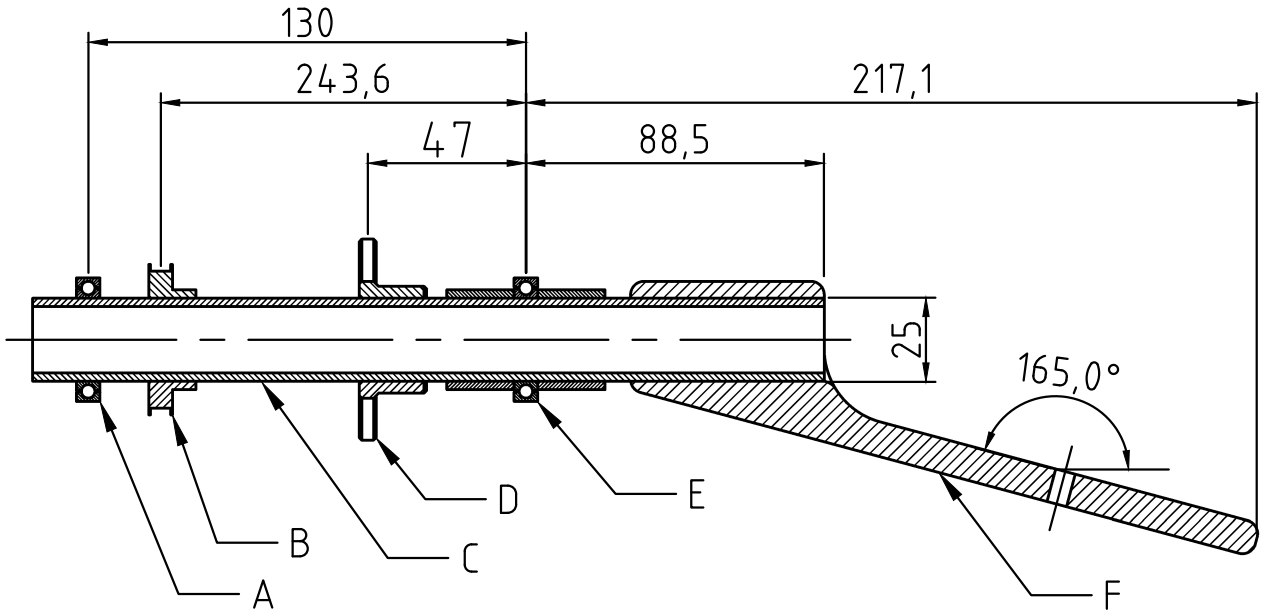


Figure 1: Main rotation axis, section view. Dimensions in mm. Drawn at 1:1 (170 mm wide).

Item	Description
A	Bearing at the drive end of the shaft
B	Toothed belt pulley (driven by the stepper)
C	Hollow shaft, 25 mm diameter
D	Index flag for the light barrier
E	Bearing with sleeve on the lidar side
F	Lidar mount, angled at 165° to the shaft axis

TODO: Confirm item descriptions, bearing type and size, shaft material and wall thickness.

The shaft is carried by two bearings 130 mm apart (Figure 1). The lidar mount extends 217.1 mm beyond the lidar-side bearing.

TODO: Insert photo of the assembled rig, e.g. `image("figures/rig_photo.jpg")`

Listing 1: Assembled rotation mechanism.

2.2.1 Drive and ratio

The total reduction from motor to lidar shaft (gearbox and belt together) was measured with the index sensor:

Quantity	Value
Microstepping	16 (3200 steps per motor revolution)
Measured steps per lidar revolution	31999 (mean of 5 revolutions)
Total reduction	10.00 : 1
Revolution-to-revolution spread	25 steps ≈ 0.28°

TODO: Gearbox ratio, pulley tooth counts, belt length and width, motor type and rated torque.

2.2.2 Belt tensioning

The motor sits on a slotted 3D-printed mount that slides on the extrusion, so the belt centre distance can be adjusted and locked.

TODO: Measured belt tension / pluck frequency, final motor mount design.

2.3 Index sensor

A light barrier detects the passage of a flag on the lidar shaft once per revolution. It sits after the belt on the lidar shaft, so skipped belt teeth appear as a changed step count per revolution. The sensor edge occurs when the lidar is 93° (8266 steps) before vertical in the scan direction.

2.4 Electronics

TODO: Teensy 4.0, MKS SERVO42C driver, BNO085 IMUs, wiring and pin table, power supply, battery.

3 Control software

Firmware for the Teensy 4.0, in `Arduino/`. The main sketch is `sketch_oct2a`.

3.1 Step generation and calibration

The firmware generates step pulses with a ramped step rate. On start-up the number of steps per lidar revolution is known from calibration (31999). It can be re-measured with the `c` command, which averages 5 revolutions between index edges.

3.2 Homing and zero position

The `h` command runs to the index edge and then moves a fixed offset to bring the lidar vertical. The offset was found by jogging the lidar with the serial commands until vertical:

$$\theta_{\text{offset}} = 93^\circ = 8266 \text{ steps}$$

3.3 Operation

Continuous rotation at 6 rpm (the lidar supports 5 to 10 rpm). Each revolution the firmware prints the step count and its deviation from the calibrated value. A persistent deviation indicates belt slip or skipped steps.

Command	Function
`c`	Calibrate steps per revolution
`h`	Home to the index edge, move offset, hold
`+ - [ ]`	Jog and adjust the offset by $\pm 1^\circ$ / $\pm 10^\circ$
`g`	Continuous rotation at 6 rpm
`x`	Stop and hold
`r`	Reverse direction

TODO: Time synchronisation between lidar packets, IMU and step counter; data output format.

4 Reconstruction and viewing software

Post-processing in `PostProcessing/` (Julia). Poses are represented with dual quaternions.

TODO: Coordinate frames, calibration of the lidar-to-IMU transform, fusion of IMU orientation with the shaft angle, point cloud reconstruction pipeline, viewer.

5 Appendix

5.1 Change log

Date	Change
2026-10-04	Index sensor, step calibration (10:1) and vertical offset (93°) measured.